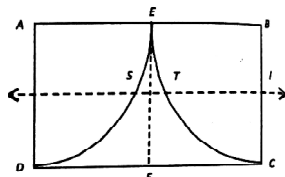


AMTI (NMTC) - 2012
BHASKARA CONTEST - JUNIOR LEVEL

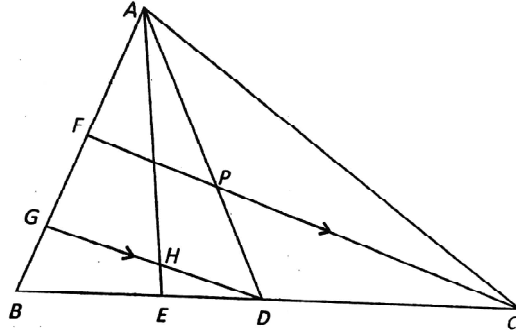
PART - A

- Two regular polygons of same number of sides have sides 40 cm and 9 cm is length. The lengths of the side of another regular polygon of the same number of sides and whose area is equal to the sum of the areas of the given polygons is (in cm)
 (A) 49 (B) 31 (C) 41 (D) 360
- If $x > y > 0$ and $\frac{x+y}{x-y} = \sqrt{2}$, the value of $\frac{x^2 + y^2}{xy}$ is
 (A) 5 (B) 4 (C) 1 (D) 6
- In rectangle ABCD, $AB = 2BC = 4$ cm E and F are midpoints of AB and CD respectively. ESD and ETC are arcs of the circles centred at A and B respectively. If the perpendicular bisector line l of EF cuts the arcs at S and T as in the diagram, then ST is equal to (in cm)



- (A) $(4 - 2\sqrt{3})$ (B) $(3 + \sqrt{3})$ (C) $(2 + 2\sqrt{3})$ (D) $(4\sqrt{3} - 2)$
- The value of 'a' for which the expression $\left\{ \left(a^{\frac{1}{4}} - a^{\frac{1}{8}} + 1 \right)^{-1} + \left(a^{\frac{1}{4}} + a^{\frac{1}{8}} + 1 \right)^{-1} - \frac{2\sqrt[4]{a} - 2}{\sqrt{a} - \sqrt[4]{a} + 1} \right\}^{-1} - 2^{\log_4 a - 2}$ takes the value 2012 is
 (A) 4048 (B) 6036 (C) 6037 (D) 4047
- C_1 and C_2 are two non-intersecting circles whose radii are in the ratio 1 : 2. A third circle cuts the smaller circle at A and B and the bigger one at C and D. AB and CD intersect at P. The ratio of the lengths of the tangents from P to the circles C_1 and C_2 is
 (A) 1:4 (B) 1:8 (C) 1:1 (D) 1:2
- a, b, c are real numbers and none of them zero and
 $E = \left(a + \frac{1}{a} \right)^2 + \left(b + \frac{1}{b} \right)^2 + \left(ab + \frac{1}{ab} \right)^2 - \left(a + \frac{1}{a} \right) + \left(b + \frac{1}{ab} \right) \left(ab + \frac{1}{ab} \right)^2$. Then E is equal to
 (A) 2012 when $a = b = 2012$ (B) 2012 when $ab = 2012$
 (C) 4 for all real values of a and b (D) 2012 for all real values of a and b
- The angles of a triangle are in the ratio 2 : 3 : 7. The length of the smallest side is 2012 cm. The radius of the circum circle of the triangle (in cm) is
 (A) 2013 (B) 2011 (C) 4024 (D) 2012
- If $a = 2012$, $b = -1005$, $c = -1007$, then the value of $\frac{a^4}{b+c} + \frac{b^4}{c+a} + \frac{c^4}{a+b} + 3abc$ is
 (A) 2012 (B) 1 (C) 0 (D) $(2012)^3$
- ABC is a triangle with $AB = 13$ cm, $BC = 14$ cm and $CA = 15$ cm. AD and BE are the altitudes from A and B to BC and AC respectively. H is the point of intersection of AD and BE. Then the ratio $\frac{HD}{HB} =$
 (A) $\frac{3}{5}$ (B) $\frac{12}{13}$ (C) $\frac{4}{5}$ (D) $\frac{5}{9}$
- For how many positive integers values of $x \leq 100$ is $(3^x - x^2)$ divisible by 5?
 (A) 16 (B) 20 (C) 24 (D) 36
- If one root of $\sqrt{a-x} + \sqrt{b+x} = \sqrt{a} + \sqrt{b}$ is 2012, then a possible value of a, b is
 (A) (2000, 2012) (B) (4024, 2012)
 (C) (1000, 1012) (D) (1012, 1000)

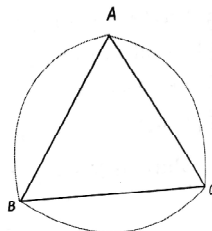
12. In the figure shown, $BD = CD$, $BE = DE$, $AP = PD$ and $DG \parallel CF$. Then $\frac{\text{area of } \triangle ADH}{\text{area of } \triangle ABC}$ is equal to



- (A) $\frac{1}{5}$ (B) $\frac{2}{9}$ (C) $\frac{3}{13}$ (D) $\frac{1}{6}$
13. If $a = 2012$, $b = 2011$, $c = 2010$ then the value of $a^2 + b^2 + c^2 - ab - bc - ca$ is
 (A) 0 (B) 2012 (C) 3 (D) 4024
14. AX and BX are two adjacent sides of a regular polygon. If $\angle ABX = \frac{1}{3} \angle AXB$, then the number of sides of the polygon is
 (A) 6 (B) 7 (C) 9 (D) 5
15. A rectangular block of dimensions $16 \times 10 \times 8$ units is painted. It is cut in to cubes of dimensions $1 \times 1 \times 1$. The number of cubes which are not painted at all is
 (A) 945 (B) 672 (C) 812 (D) 796

PART - B

1. The value of $\sqrt[3]{5 + 2\sqrt{13}} + \sqrt[3]{5 - 2\sqrt{13}}$ is = _____.
2. Triangle ABC is equilateral of side length 8 cm. Each arc shown in the diagram is an arc of a circle with the opposite vertex of the triangle as its centre. The total area enclosed within the entire figure shown (in cm^2) _____.



3. If a, b, c, d , satisfy the equations $a + 7b + 3c + 5d = 0$, $8a + 4b + 6c + 2d = -16$, $2a + 6b + 4c + 8d = 16$, $5a + 3b + 7c + d = -16$ then the value of $(a + d)(b + c)$ = _____.
4. Lines $L_1, L_2, L_3, \dots, L_{20}$ are distinct. All the lines L_4, L_8, L_{12}, L_{16} and L_{20} are parallel. All the lines $L_1, L_5, L_9, L_{13}, L_{17}$ pass through a given point A, the maximum number of points of intersection of these 20 lines is _____.
5. x, y, z are real numbers such that $(x + y)^2 = 16$, $(y + z)^2 = 36$, $(z + x)^2 = 81$, $x + y + z > 3$. The number of possible values of $(x + y + z)$ is _____.
6. ABCD is a square. A line AX meets the diagonal BD at X and $AX = 2012$ cm. the length of CX (in cm) is _____.
7. A two digit number is 6 times the sum of its digits. The number formed by interchanging the digits is k times the sum of the digits. Then values of k is _____.
8. O is the centre of a circle of radius 15 cm. M is a point at a distance of 5 cm from O. AMB is any chord of the circle through M, then the value of $AM \times MB$ is _____.
9. The combined age of a man and his wife is six times the combined ages of their children. Two years ago their united ages were ten times the combined ages of their children. Six years hence their combined age will be three times the combined age of the children. The number of children they have is _____.
10. A two digit number is less than the sum of the squares of its digits by 11 and exceeds twice the product of its digits by 5. The two digit number is _____.

- 11. An isosceles trapezoid is circumscribed about a circle of radius 2 cm and the area of the trapezoid is 20 cm². The equal sides of the trapezoid have length _____.
- 12. A triangle has sides with lengths 13 cm, 14 cm, 15 cm. A circle whose centre lies on the longest side touches the other two sides. The radius of the circle is (in cm) _____.
- 13. The sum of the roots of the equation $x^3\sqrt{x^2}=(\sqrt{x})^x$ is _____.
- 14. ABC and ADE are two secants of a circle of radius 3 cm. A is at a distance of 5 cm from the centre of the circle. The secants include an angle of 30°. The area of the ΔACE is 10 cm². Then the area of the ΔADB (in cm²) is _____.
- 15. The value of x which satisfies the equation $5^2 \cdot 5^4 \cdot 5^6 \dots 5^{2x} = (0.04)^{-28}$ is _____.
